Applied Mathematics for Chemists
(화학수학)

Lecture 2

Review of Calculus

Today's Class

- Operations: powers and roots
- Numbers
  - Complex numbers
- Functions
  - Logarithms and exponentials
  - Trigonometric functions
- Calculus
  - Differentiation
  - Integration

Numerical Mathematics

- Powers (거듭제곱)

$$x^2 = x \times x, \qquad x^3 = x \times x \times x,$$

$$x^n = x \times x \times \cdots \times x$$

- Roots (제곱근)

$$(\sqrt{x})^2 = x, \qquad (\sqrt[3]{x})^3 = x,$$

$$(\sqrt[n]{x})^n = x$$

Exponents (지수)

n in x^n = exponent of x

- Negative exponents

$$x^{-1} = \frac{1}{x}, \qquad x^{-3} = \frac{1}{x^3}$$

- Fractional exponents

$$x^{1/2} = \sqrt{x},\ x^{1/3} = \sqrt[3]{x}$$

Properties of Exponents

- Property 1

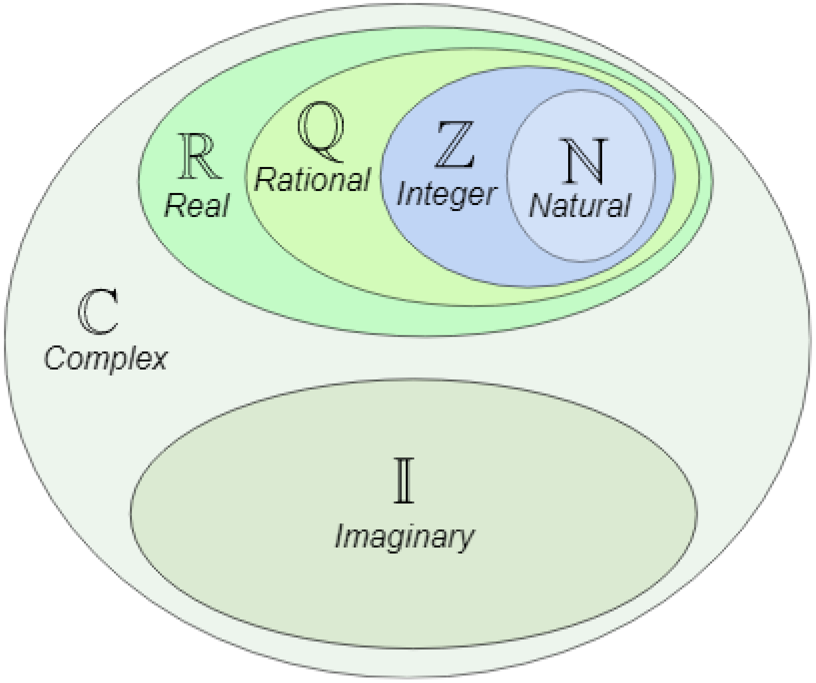
$$x^a x^b = x^{a+b}$$

- Property 2

$$(x^a)^b = x^{ab}$$

Not in Mortimer

Numbers



(Image: https://www.mathsisfun.com/sets/number-types.html)

Complex Numbers (복소수)

- Imaginary unit (허수 단위)

$$i = \sqrt{-1}$$

- Complex number (a, b: real)

$$c = a + ib$$

where $a = Re(c)$ is the real part of c and $b = Im(c)$ is the imaginary part of c.

Operations with Complex Numbers

If $c_1 = a_1 + ib_1$ and $c_2 = a_2 + ib_2$,

- $c_1 + c_2 = (a_1 + a_2) + i(b_1 + b_2)$

- $c_1 c_2 = (a_1 + ib_1)(a_2 + ib_2)$

$$= a_1 a_2 + i(a_1 b_2 + a_2 b_1) + i^2 b_1 b_2$$

$$= (a_1 a_2 - b_1 b_2) + i(a_1 b_2 + a_2 b_1)$$

Related Quantities

If a complex number is given as $c = a + ib$, its ⋯

- Complex conjugate (켤레복소수)

$$c^* = \bar{c} = (a + ib)^* = a - ib$$

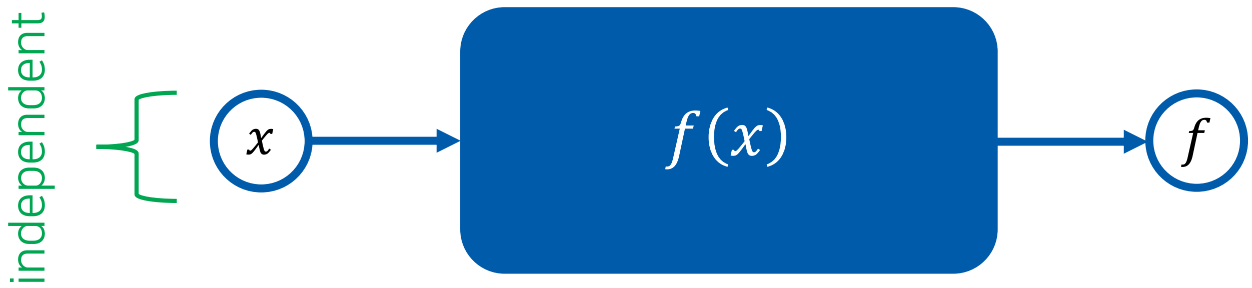
- Magnitude (크기)

$$|c| = \sqrt{cc^*} = \sqrt{a^2 + b^2}$$

- Reciprocal (역수)

$$c^{-1} = \frac{a - ib}{a^2 + b^2} = \frac{c^*}{|c|^2}$$

Function (함수)



Example: $f(x) = x^2 + 2x$

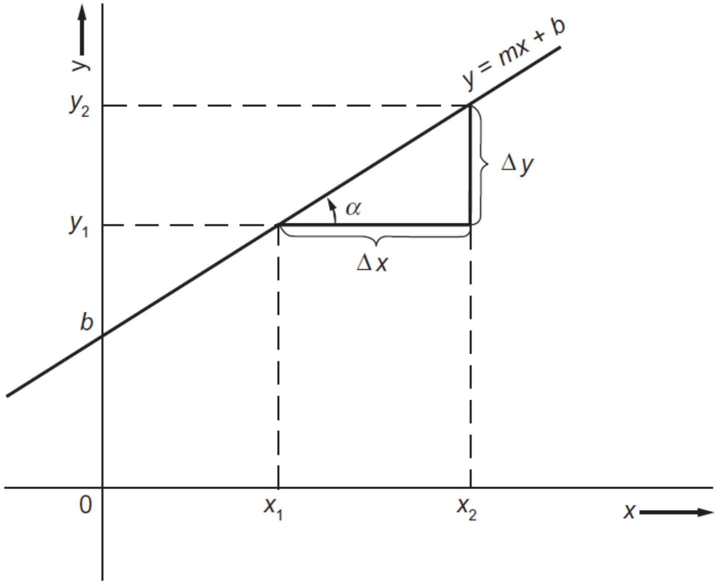
$$f(1) = 1^2 + 2 \times 1 = 3$$

$$f(-1) = (-1)^2 + 2 \times (-1) = -1$$

$$\vdots$$

Linear Functions (선형함수)

$$y = mx + b$$



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 2.2)

Polynomial Functions (다항함수)

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots$$

- Quadratic functions

$$y = ax^2 + bx + c$$

- Cubic functions

$$y = ax^3 + bx^2 + cx + d$$

Logarithms (로그함수)

$$x = a^y \Leftrightarrow y = \log_a x$$

- Common logarithms (상용로그): $a = 10$

- Natural logarithms (자연로그): $a = e$, $\log_e x = \ln x$

$$e = \lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n = 2.718\ldots$$

Properties of Exponents and Logarithms

Exponent fact	Logarithm fact
$a^0 = 1$	$\log_a(1) = 0$
$a^{1/2} = \sqrt{a}$	$\log_a(\sqrt{a}) = \frac{1}{2}$
$a^1 = a$	$\log_a(a) = 1$
$a^{x_1} a^{x_2} = a^{x_1+x_2}$	$\log_a(y_1 y_2) = \log_a(y_1) + \log_a(y_2)$
$a^{-x} = \frac{1}{a^x}$	$\log_a\left(\frac{1}{y}\right) = -\log_a(y)$
$\frac{a^{x_1}}{a^{x_2}} = a^{x_1-x_2}$	$\log_a\left(\frac{y_1}{y_2}\right) = \log_a(y_1) - \log_a(y_2)$
$(a^x)^z = a^{xz}$	$\log_a(y^z) = z\log_a(y)$
$a^\infty = \infty$	$\log_a(\infty) = \infty$
$a^{-\infty} = 0$	$\log_a(0) = -\infty$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Table 2.2)

Exponentials (지수함수)

$$y = e^{bx} = \exp(bx)$$

- Conversion of $y = a^{bx}$

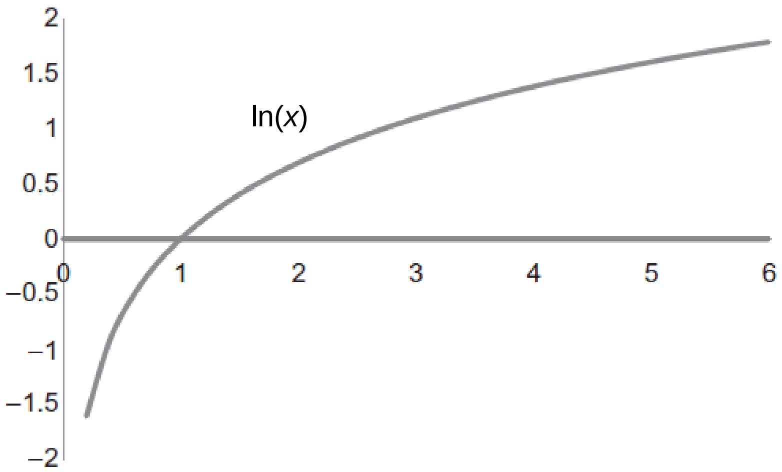
Assume that $a = e^n$

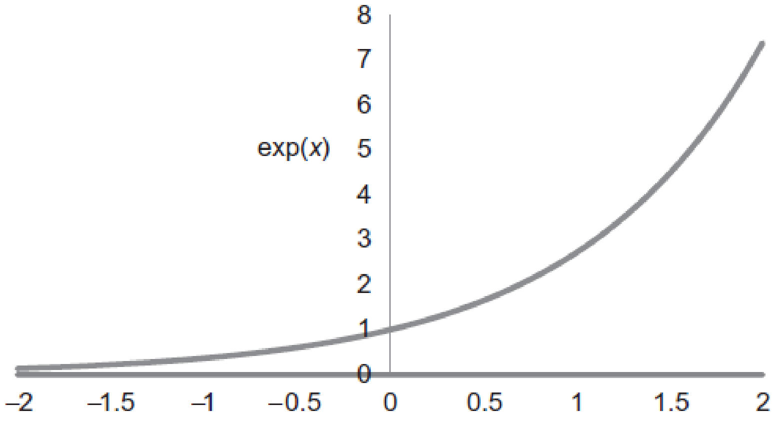
This gives $n = \log_e a = \ln a$

Substitution leads to

$$y = (e^n)^{bx} = e^{nbx} = e^{bx\ln a}$$

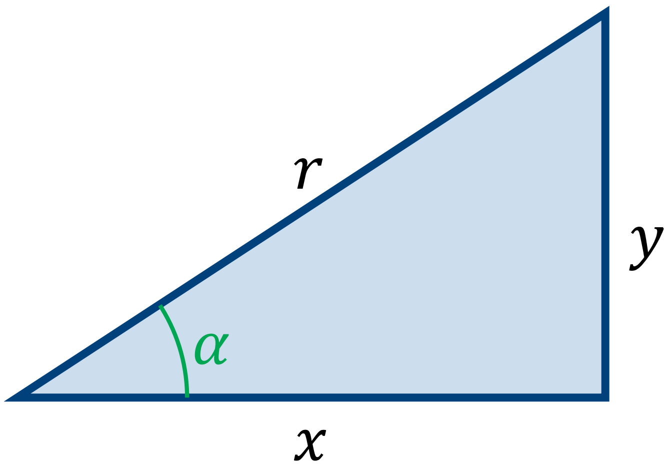
Logarithms and Exponentials





(Image: Mortimer, *Mathematics for Physical Chemistry*, 4^{th} ed., Figures 2.4 and 2.5)

Trigonometric Functions (삼각함수)



$$\sin\alpha = \frac{y}{r}, \qquad \csc\alpha = \frac{1}{\sin\alpha} = \frac{r}{y}$$

$$\cos\alpha = \frac{x}{r}, \qquad \sec\alpha = \frac{1}{\cos\alpha} = \frac{r}{x}$$

$$\tan\alpha = \frac{y}{x}, \qquad \cot\alpha = \frac{1}{\tan\alpha} = \frac{x}{y}$$

Trigonometric Identities (삼각함수 항등식)

$$\sin\alpha = -\sin(-\alpha)$$

$$\cos\alpha = \cos(-\alpha)$$

$$\tan\alpha = -\tan(-\alpha)$$

$$\sin\alpha = \sin(\alpha + 2\pi) = \sin(\alpha + 4\pi) = \cdots$$

$$\cos\alpha = \cos(\alpha + 2\pi) = \cos(\alpha + 4\pi) = \cdots$$

$$\sin^2\alpha + \cos^2\alpha = 1$$

Trigonometric Functions

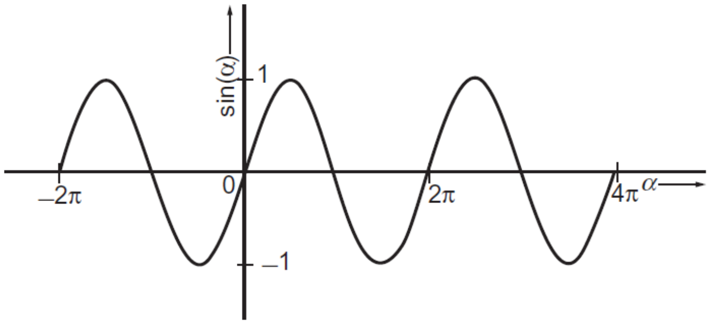


FIGURE 2.7 The sine of an angle.

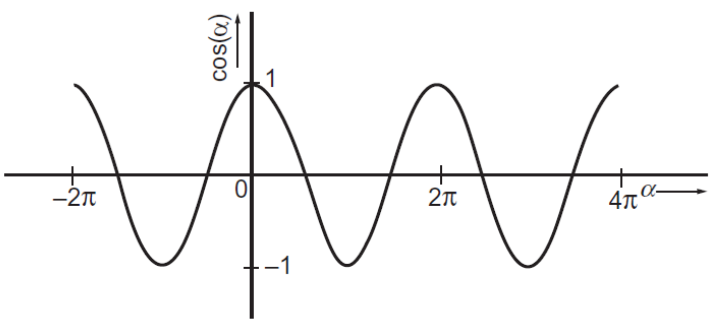


FIGURE 2.8 The cosine of an angle.

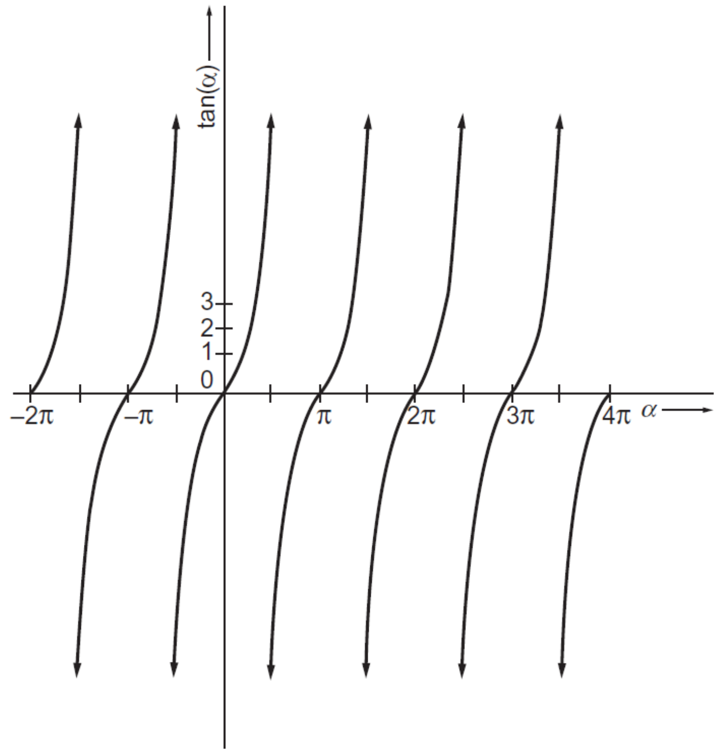


FIGURE 2.9 The tangent of an angle.

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figures 2.7, 2.8, and 2.9)

Inverse Trigonometric Functions

(역 삼각함수)

$$y = \sin x \iff x = \arcsin y = \operatorname{asin} y = \sin^{-1} y$$

$$y = \cos x \iff x = \arccos y = \operatorname{acos} y = \cos^{-1} y$$

$$y = \tan x \iff x = \arctan y = \operatorname{atan} y = \tan^{-1} y$$

Euler's Formula

$$e^{ix} = \cos x + i \sin x$$

- Application of Euler's formula

$$e^{ix} = \cos x + i \sin x$$

$$e^{-ix} = \cos x - i \sin x$$

Addition and subtraction leads to

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}, \qquad \sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

Hyperbolic Trigonometric Functions

(쌍곡 삼각함수)

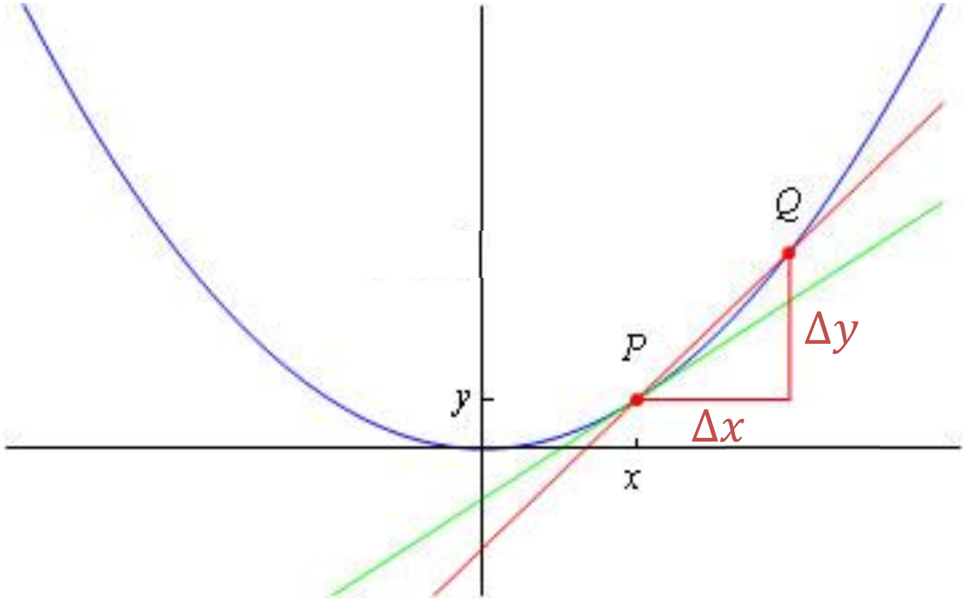
$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}$$

$$\tanh x = \frac{\sinh x}{\cosh x}, \qquad \coth x = \frac{1}{\tanh x}$$

$$\operatorname{sech} x = \frac{1}{\cosh x}, \qquad \operatorname{csch} x = \frac{1}{\sinh x}$$

Differentiation

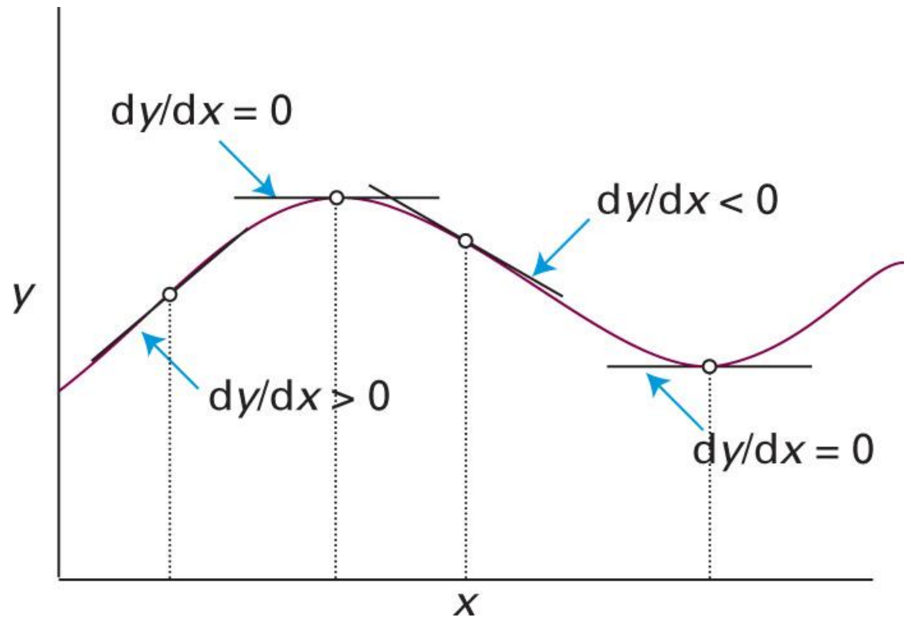
"The slope of a function"



$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \frac{\mathrm{d}y}{\mathrm{d}x}$$

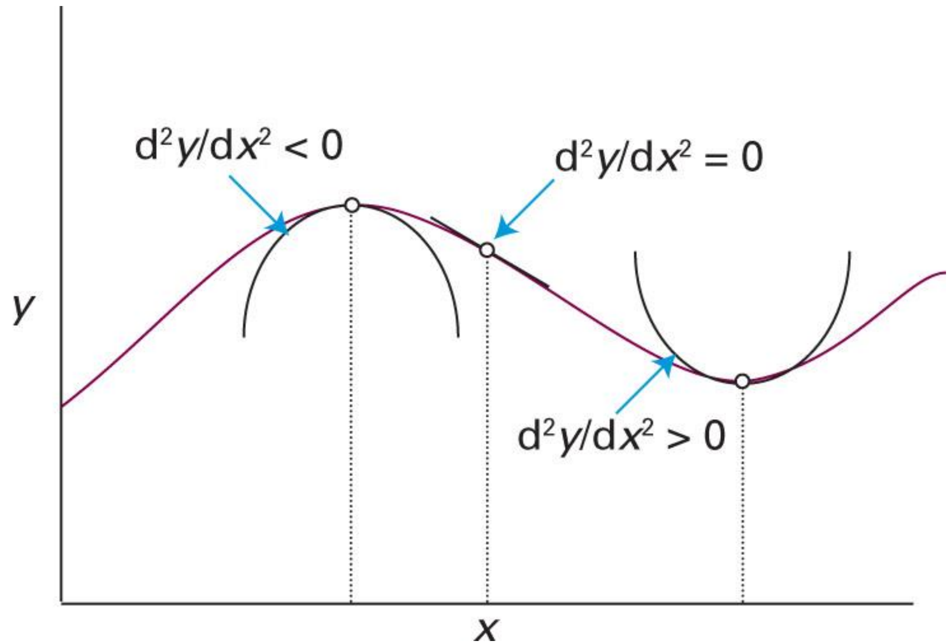
(Image: http://wwwf.imperial.ac.uk/metric/metric_public/differentiation/index.html)

First Derivative: Slope



(Image: Atkins *et al*., *Atkins' Physical Chemistry*, 11th ed., the chemist's toolkit 5, Sketch 1)

Second Derivative: Sharpness



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., the chemist's toolkit 5, Sketch 2)

Some Examples of Derivatives

Function, $y = y(x)$	Derivative, $dy/dx = y'(x)$
ax^n	anx^{n-1}
ae^{bx}	abe^{bx}
a	0
$a \sin(bx)$	$ab \cos(bx)$
$a \cos(bx)$	$-ab \sin(bx)$
$a \ln(x)$	a/x
$a \tan(x)$	

Check Appendix D!

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Table 6.1)

Derivative Identities (1)

- If y is a function of x and c is a constant,

$$\frac{\mathrm{d}(y+c)}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}x}, \qquad \frac{\mathrm{d}(cy)}{\mathrm{d}x} = c\,\frac{\mathrm{d}y}{\mathrm{d}x}$$

- If y and z are both functions of x,

$$\frac{\mathrm{d}(y+z)}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{\mathrm{d}z}{\mathrm{d}x}$$

$$\frac{\mathrm{d}(yz)}{\mathrm{d}x} = z\,\frac{\mathrm{d}y}{\mathrm{d}x} + y\,\frac{\mathrm{d}z}{\mathrm{d}x}$$

Derivative Identities (2)

- If y and z are both functions of x,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{z}\right) = \frac{1}{z^2}\left(z\frac{\mathrm{d}y}{\mathrm{d}x} - y\frac{\mathrm{d}z}{\mathrm{d}x}\right)$$

- If u is a function of x and f is a function of u,

$$\frac{\mathrm{d}f}{\mathrm{d}x} = \frac{\mathrm{d}f}{\mathrm{d}u}\frac{\mathrm{d}u}{\mathrm{d}x}$$

(Chain rule)

Example 6.6

Find $\mathrm{d}P/\mathrm{d}T$ if $P(T) = ke^{-Q/T}$.

Let $u = -Q/T$. Then, $P(u) = ke^{u}$. From the chain rule,

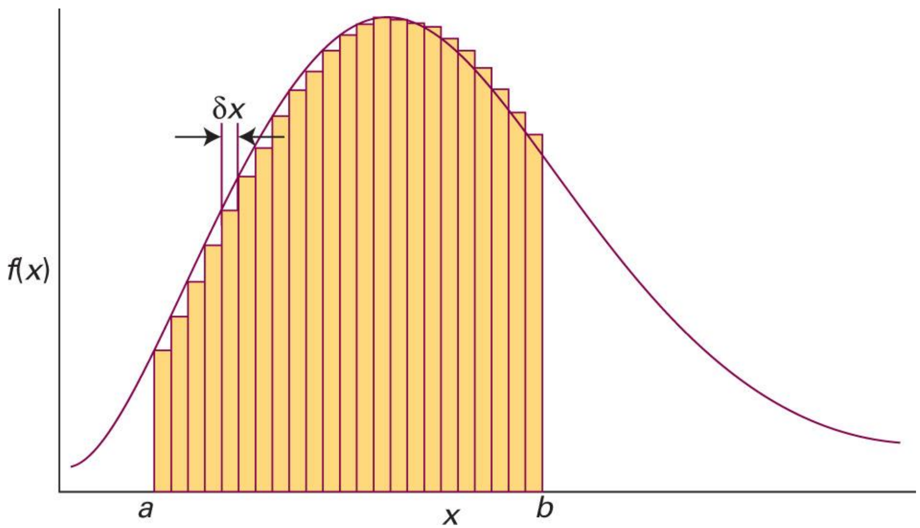
$$\frac{\mathrm{d}P}{\mathrm{d}T} = \frac{\mathrm{d}P}{\mathrm{d}u}\frac{\mathrm{d}u}{\mathrm{d}T} = \frac{\mathrm{d}(ke^{u})}{\mathrm{d}u} \times \frac{\mathrm{d}}{\mathrm{d}T}\left(-\frac{Q}{T}\right) = ke^{u} \times \frac{Q}{T^2}$$

Therefore,

$$\frac{\mathrm{d}P}{\mathrm{d}T} = \frac{kQ}{T^2}e^{-Q/T}$$

Integration

"The areas under curves"



$$\lim_{\delta x \to 0} \sum_i f(x_i)\delta x = \int_a^b f(x)\,\mathrm{d}x$$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., the chemist's toolkit 4, Sketch 1)

Indefinite Integrals (부정적분)

Consider an integral

$$\int_a^{x'} f(x)\,\mathrm{d}x = F(x') + C$$

constant of integration (적분상수)

where x' is a variable and a is an arbitrary constant.

Generally, it is written as

$$\int f(x)\,\mathrm{d}x = F(x) + C$$

Calculation of Indefinite Integrals

Differentiation $\xleftrightarrow{\text{Inverse}}$ Integration

When evaluating the integral $F(x) = \int f(x)\mathrm{d}x$,

find a function $F(x)$ whose derivative is $f(x)$:

$$f(x) = \frac{\mathrm{d}F(x)}{\mathrm{d}x}$$

Some Examples of Derivatives

Integration

Function, $y = y(x)$	Derivative, $dy/dx = y'(x)$
ax^n	anx^{n-1}
ae^{bx}	abe^{bx}
a	0
$a \sin(bx)$	$ab \cos(bx)$
$a \cos(bx)$	$-ab \sin(bx)$
$a \ln(x)$	a/x
$a \tan(x)$	

Check Appendix E!

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Table 6.1)

Calculation of Definite Integrals

If the indefinite integral of $f(x)$ is $F(x)$,

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a)$$

No constant of integration!

Techniques of Integration

Method of substitution (변수치환법)

Integration by parts (부분적분법)

Method of partial fractions (부분분수법)

Method of Substitution

1. Define a new variable
2. Calculate its infinitesimal change and the limits
3. Rewrite the integral in terms of the new variable

Example 7.9

Evaluate the integral

$$\int_0^{\infty} x e^{-x^2}\, \mathrm{d}x.$$

Let $u = x^2$. Then,

$$\mathrm{d}u = 2x\, \mathrm{d}x$$

$$x = 0 \rightarrow u = 0$$

$$x = \infty \rightarrow u = \infty$$

Hence, substitution leads to

$$\int_0^{\infty} \frac{1}{2} e^{-u}\, \mathrm{d}u = \left[-\frac{1}{2} e^{-u}\right]_0^{\infty} = \frac{1}{2}$$

Integration by Parts

If u and v are both functions of x,

$$\frac{\mathrm{d}(uv)}{\mathrm{d}x} = v\frac{\mathrm{d}u}{\mathrm{d}x} + u\frac{\mathrm{d}v}{\mathrm{d}x}$$

Integrating both sides,

$$uv = \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x + \int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x + C$$

which gives

$$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x + C$$

Integration by Parts

For indefinite integrals,

$$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\mathrm{d}x = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\mathrm{d}x + C$$

For definite integrals,

$$\int_a^b u\frac{\mathrm{d}v}{\mathrm{d}x}\mathrm{d}x = uv\Big|_a^b - \int_a^b v\frac{\mathrm{d}u}{\mathrm{d}x}\mathrm{d}x$$

Example 7.11

Find the definite integral

$$\int_0^{\pi} x \sin x \, \mathrm{d}x.$$

Let $u = x$ and $\mathrm{d}v/\mathrm{d}x = \sin x$. Then, $v = -\cos x$.

$$\int_0^{\pi} x \sin x \, \mathrm{d}x = \int_0^{\pi} u \frac{\mathrm{d}v}{\mathrm{d}x} \mathrm{d}x = uv\Big|_0^{\pi} - \int_0^{\pi} v \frac{\mathrm{d}u}{\mathrm{d}x} \mathrm{d}x$$

Hence, substitution leads to

$$x(-\cos x)\Big|_0^{\pi} - \int_0^{\pi} (-\cos x) \, \mathrm{d}x = \pi + (\sin x)\Big|_0^{\pi} = \pi$$

Method of Partial Fractions

If $P(x)$ and $Q(x)$ are polynomials, and $Q(x)$ can be factorized as

$$Q(x) = (a_1 x + b_1)(a_2 x + b_2) \cdots (a_n x + b_n),$$

$P(x)/Q(x)$ can decomposed as

$$\frac{P(x)}{Q(x)} = \frac{A_1}{a_1 x + b_1} + \frac{A_2}{a_2 x + b_2} + \cdots + \frac{A_n}{a_n x + b_n}$$

Simpler combination of fractions

Example 7.12

Decompose the integral

$$\int \frac{6x-30}{x^2+3x+2}\,\mathrm{d}x.$$

The denominator is decomposed as $x^2+3x+2=(x+2)(x+1)$. Thus,

$$\frac{6x-30}{x^2+3x+2}=\frac{A_1}{x+2}+\frac{A_2}{x+1}$$

Multiplying both sides by $(x+2)(x+1)$,

$$6x-30=A_1(x+1)+A_2(x+2)$$

$$6x-30=(A_1+A_2)x+(A_1+2A_2)$$

Since $A_1+A_2=6$ and $A_1+2A_2=-30$, $A_1=42$ and $A_2=-36$. Hence,

$$\int \frac{6x-30}{x^2+3x+2}\,\mathrm{d}x=\int \frac{42}{x+2}\,\mathrm{d}x-\int \frac{36}{x+1}\,\mathrm{d}x.$$